

VGIC Informal Feedback on Dominion VPP Proposal

Background

On May 2nd, Governor Youngkin signed the Community Energy Act (HB 2346 / SB 1100) into law, directing Dominion Energy Virginia, the state's largest utility, to:

- Propose a 450 MW VPP pilot program by December 1, 2025
 - Propose a program tariff for residential and C&I customers by November 15, 2026
 - Participants may enroll directly with utility or through a third-party aggregator
 - By December 1, 2027, propose a V2G school bus program

Dominion currently offers:

- A \$40/yr managed charging program for eligible Level 2 chargers (ChargePoint, Emporia, EVO, and Wallbox)
- A whole-home off-peak charging rate for interested customers
- Dual participation in the two is not permitted

View Dominion's VPP proposal here: <https://www.dominionenergy.com/virginia/save-energy/virtual-power-plant>.

Dominion Feedback Form and Proposed VGIC Response

Dominion's questions are listed below in **bold**; VGIC's responses are listed below each bolded question.

What benefits do you see a VPP bringing to customers, businesses, and communities?

Given the significant anticipated load growth expected in Dominion's service territory due to new data centers, VPPs are one of the only tools available to Dominion to unlock affordability and reliability benefits that, in turn, accrue to all customers, including non-participating ratepayers. Meanwhile, participating customers benefit from bill savings and/or credits and, for customers with on-site generation and/or storage, resilience during outages. These benefits are particularly meaningful for residential bidirectional charging solutions, which Dominion should deem eligible for the VPP pilot in recognition of the significant interest from automakers, including those in VGIC's 43 members representing automakers, EV charging companies, EV aggregators, charging station installers and O&M providers, and EV charging site developers.

What should Dominion Energy Virginia seek to demonstrate and/or learn from this pilot?

Dominion should ensure that residential bidirectional charging systems are supported through this pilot, as bidirectional-capable vehicle offerings from Ford, GM, Tesla, Kia, Honda, Nissan, Mercedes-Benz, BMW, Stellantis, Hyundai, Rivian, Lucid, and Toyota are either currently available or will be released in the next few model years, and the corresponding charging equipment from Wallbox, dcbel, ChargePoint, Emporia, Autel, Enphase, and several others are in the same status.

Additionally, Dominion should seek to develop and implement strategies needed to facilitate its planned scale-up from limited pilot to mass-market offerings. Specifically, Dominion should develop and implement customer enrollment and retention strategies, including specific strategies targeting residential EV drivers and non-residential fleets interested in unidirectional managed charging solutions or grid-parallel bidirectional charging systems (sometimes referred to as “vehicle-to-grid” systems).

Dominion must also develop and implement strategies to facilitate opportunities for aggregators and third-party technology providers to enable effective customer response. Notably, this is a statutory requirement of HB 2346 / SB 1100, and must be implemented in the VPP pilot. This access should be non-discriminatory, meaning aggregators should have comparable ability to enroll customers as Dominion. Given Dominion will likely serve as both platform operator and competitor to aggregators, Dominion should explicitly propose guardrails against utility self-preferencing. The relevant data access protocols and platforms, APIs or third-party integration platforms, and any commercial and legal agreements must be developed and implemented in this pilot to yield key learnings that will facilitate later scale-up to mass-market VPP offerings. Additionally, Dominion should develop and implement strategies to test the efficacy of different payment structures or levels.

Dominion should consider implementing flexible service connection offerings that critically inform initial site design and construction, especially for larger commercial and industrial sites that can leverage flexible connection agreements to avoid, defer, or serve as a bridge to lengthy utility-side upgrade processes. These flexible service connection offerings not only accelerate energization timelines, but also avoid or defer system costs and, in doing so, can support affordability for all customers, not just those that elect flexible connection arrangements. Dominion should consider implementing a shared savings model for these customers, wherein a share of the avoided or deferred costs are provided upfront or over time as part of VPP participation.

Above all else, the VPP pilot must focus on how to make VPPs successful from the perspective of customer participation (e.g., understanding what drives strong customer engagement in a VPP), value/metrics (e.g., illustrating real-world benefits and optimal program design at scale), and equity (e.g., demonstrating broad/equitable participation).

What metrics should be used to measure pilot outcomes?

Metrics should always be explicitly tied to the Goals and Outcomes of the pilot. Goals provide the highest-level orientation for that the VPP pilot aims to achieve. For example, compliance with HB 2346 / SB 1100, increased system reliability, and affordability can inform the identification of appropriate Goals. Outcomes are more specific factors that derive from Dominion's operations and business decisions. Outcomes are observable, whether qualitatively or quantitatively. Metrics are the way Outcomes are measured. Each Metric should include (a) specific data definitions, (b) precise formula used to quantify each metric, (c) data collection and analysis techniques, including identifying the entity responsible for collecting and/or reporting the data, (d) requirements for measurement and reporting, and (e) verification techniques.

As for the specific Metrics to be used, VGIC recommends that, in addition to the Goals-Outcomes-Metrics framework, all relevant inputs for Lawrence Berkeley National Lab's Benefit-Cost Analysis Guide for Managed Charging Programs be collected from the VPP pilot. The guide can be found here: emp.lbl.gov/publications/best-practices-guide-benefit-cost. Critically, the guide includes details on baselining methodologies to support the measurement of VPP participation relative to business-as-usual operations. Collecting these inputs can inform the cost-effectiveness testing for future, large-scale VPP programs.

Any metrics needed to shape Dominion's integrated resource planning process must be collected. If these metrics are not incorporated into the IRP process and the resulting selection of resources and investment decisions, then the benefits from VPP implementation will remain hypothetical. By ensuring VPP data collection is explicitly tied to IRP modeling, Dominion can unlock reliability and affordability benefits in practice, not just in theory.

As detailed below regarding barriers related to complexity, VGIC does not recommend that the initial VPP pilot seek to integrate resources into the PJM market. However, Dominion's evaluation and measurement of the VPP pilot should include recommendations to evolve the VPP offering to align with the vision of FERC Order 2222, including identifying and proposing solutions to any dual participation, double counting, interconnection tariff, site telemetry, performance penalties, and/or foundational cross-jurisdictional policy issues.

Given the apparent increased risk of extreme weather events, cybersecurity attacks, and physical security attacks, VGIC recommends Dominion estimate the value of resilience that the program enables. To the extent that the VPP pilot encourages adoption of local and customer-sited generation and/or energy storage, the resulting resilience benefits for participants should be accounted for in assessing the pilot. While this is hard to quantify, metrics like business interruption cost/lost revenues, lost assets, healthcare costs, and/or individual customers' willingness to pay for backup power can be collected to derive a value of lost load.

Most importantly, all metrics and data should be publicly reported on a predictable timeline. This should include, at a minimum, interim and final reports. Dominion and/or the SCC should host stakeholder engagement opportunities and welcome questions and recommended enhancements from interested stakeholders. This transparency is critical to ensure the VPP pilot does not become a siloed experiment confined to Dominion's internal stakeholders.

Do you have any concerns about how a VPP may impact customers, business, and communities?

Yes. Insufficient savings and/or bill credit opportunity and abrupt program disruption can both result in a poor customer experience. Additionally, prohibiting customers with grid-parallel generation and/or energy storage, including residential bidirectional charging systems, from participating in the program, as well as prohibiting net exporting systems (as opposed to non-export, grid-parallel systems), will negatively impact the customer experience. A lack of clear guidance to customers and aggregators on what is needed of them to apply and participate in the program will restrict the opportunities for customers to enroll, participate in the VPP, and, in turn, support the reliability and affordability challenges facing Dominion and its customers.

Additionally, Dominion should deem the widest range of technological pathways eligible for enrollment and participation. For example, a customer should be given the choice to participate in the VPP pilot via EV and/or EV charger, and the list of eligible devices should be inclusive of a variety of options. Limiting the enrollment and participation pathways to an overly limited number or type of devices will severely restrict participation.

Relatedly, Dominion should incentivize or otherwise promote interoperability and devices using open standards to ensure long-term scalability and customer satisfaction.

What would motivate you (or your customers/constituents) to participate in a VPP?

Sufficient savings and/or bill credit opportunity is critical to encourage customers and aggregators to participate in a VPP. National Grid and Eversource's Connected Solutions –

Daily Dispatch option is an example of a cost-effective, technology-agnostic, and aggregator-friendly VPP program that sufficiently attracts customers. The program is underpinned by high ISO/RTO costs that utilities would otherwise be subject to. Xcel Energy Colorado's recently filed VPP Settlement provides another example structure and magnitude of performance-based compensation that could motivate customers and aggregators to participate in a VPP.

Ensuring eligibility for residential bidirectional charging systems is critical to motivating participation from the automotive industry actors, including the automakers themselves, third-party aggregators such as ChargeScape, WeaveGrid, EnergyHub, ev.energy, Kaluza, Optiwatt, Bidirectional Energy, Nuvve/Fermata, and The Mobility House, and the charging companies including ChargePoint, Emporia, dcbel, Wallbox, and Enphase.

What barriers might prevent participation (e.g., cost, complexity, trust)?

(1) Insufficient savings and/or credit opportunity, (2) limiting VPP pilot compensation to load-only and/or non-export resources, and/or (3) imposing onerous interconnection requirements for grid-parallel generation/storage resources, including residential bidirectional V2G charging equipment, will undoubtedly prevent participation.

Additionally, too much friction in the enrollment process, generally, will limit participation. Streamlined portals and partner and installer trainings will be key to the success of the VPP pilot.

Regarding complexity, VGIC recommends that the initial VPP pilot be designed and implemented by Dominion Energy as a retail utility program offering in which customer resources are not integrated into the PJM wholesale market. While in the long run, VGIC believes that wholesale market integration, as envisioned by FERC Order 2222, is a critical step to advance VPP, we recommend that Dominion focus on its own implementation of this as a demand-side management customer program, rather than a ISO/RTO participation pathway.

Above all else, the lack of a consistent, predictable program opportunity will preclude customer and aggregator participation. This is particularly true for resources and/or product development capabilities that require significant upfront investment. This has been borne out across the country, where compelling two- or three-year VPP pilots see little participation and, when asked, the critical enabling technology providers site a lack of certainty in the long-term stability of the opportunity as a primary barrier.

How can Dominion Energy Virginia ensure that low-income or underserved households are able to access VPP benefits?

First, the VPP should be at significant scale and compensation levels to attract customer participation both inside and outside of low-income and underserved communities. The underlying reliability and, most importantly, affordability benefits of a VPP will accrue to all customers, including customers not participating in VPPs. Low-income and underserved households are hit hardest by increased rates. Through earnest and effective VPP implementation, Dominion can support these most burdened customers by lowering their rates on average.

Second, the VPP should ensure low-income and underserved households can participate in the VPP, including customers in both single-family and multi-family residences. This can be done through adders to performance-based compensation, upfront incentives to offset equipment costs and related infrastructure, on-bill financing and other alternative financing options, and/or targeted customer marketing, education, and outreach.

Third, as it relates to eligible EV technologies, customer should be enabled to enroll through their vehicle, charger, or other device. This will enable fair access to the program, as customers in single-family homes, multi-family homes, and rural communities have varying degrees of access to Wi-Fi, broadband, or other means of coordinating VPP enrollment and participation. Should Dominion choose an active, event-based program design for its VPP pilot, customers should always be permitted to opt-out of a given event. These opt-out rights are particularly important for EV drivers, as EVs are primarily used as a mobility resource.

What role should local governments or community groups play in a VPP?

Dominion should engage local government and community groups to support customer marketing, education, and outreach. Additionally, Dominion should engage with local authority having jurisdictions responsible for permitting to streamline the local electrical permitting process, especially for customer-sited generation and energy storage. This is particularly critical for emerging solutions like bidirectional EV charging systems, which may be new for many AHJs.

Please provide any additional feedback or recommendations that you would like Dominion Energy Virginia to consider for its VPP.

Dominion should ensure that proposals for future VPP pilot extension/expansion are brought to stakeholders early and filed with its regulator long before any customer cap, budget cap, or sunset date for the initial VPP pilot are reached. If there is a disruption in program authorization and implementation, customers may never come back to the extended/expanded offering and aggregators may look to other markets to engage with utilities offering more consistent and predictable opportunities.

Dominion should ensure its interconnection rules and processes are modernized to accommodate newer types of IEEE 1547-2018-compliant and UL 1741 SB or SC-certified DERs, namely bidirectional charging systems. The proper forms, portals, and guidance should be clear and readily available to installers and site developers.

Lastly, VGIC reiterates that Dominion should include residential V2G systems as eligible for the VPP pilot. The list of commercially-available and validated products has grown significantly over the previous 12-18 months and will continue to do so in the coming 6-12 months, as several new charging systems gain UL certification, a critical milestone and gating item for commercial launch of these systems. VGIC strongly recommends Dominion seriously consider how to accommodate these resources and, in turn, position itself as a national leader on V2G. VGIC offers itself as a resource in the VPP design process.